Date: Tuesday, May 05th, 2026

Time: 1:30 AM to 04:00 PM

Lab Instructor(s):

Ms. Saira Ayub.

Total Marks: 100

Total Questions: 04

Paper Type: B

[redacted]

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Attempt all the questions!

Important Instructions:

- Please submit a Word file that includes all SQL queries along with screenshots of the output for each query.
- Rename your final submission file as: DB_Systems_Final_Lab_Exam_Your-Name_Your-Roll-No

**LO # 02: Design relational data models using Entity–Relationship Model and implement database structures
integrity constraints using SQL**

Question # 01: **[Weightage: 10 | Marks: 20 | Time: 40 minutes]**

The Smart Agriculture Monitoring and Supply Chain System manages farming activities, crop production, resources, and distribution across multiple farming zones. Each farming zone has a unique zone code, name, climate type, soil quality, and total cultivated area, and it can contain multiple farms.
The system stores farmer information, including farmer ID, name, experience, and contact details. Farmers can manage multiple zones, and each zone can have multiple farmers, along with the start date of their supervision.
Crops are also tracked with details such as crop ID, name, season, and growth duration. A zone can grow multiple crops, and the same crop can be grown in different zones. Agricultural equipment is managed with details like equipment ID, type, purpose, and status. Each equipment item belongs to one zone, while a zone can have multiple equipment items.
Harvest records are maintained with harvest ID, date, quantity, and quality grade. Each record is linked to one crop and one zone, but multiple records can exist over time.
The system also manages distribution centers, including center ID, location, capacity, and manager. Crops from different zones can be sent to multiple centers, and each center can receive crops from multiple zones.
The system focuses on farming operations, crop management, equipment usage, harvest tracking, farmer assignments, and distribution logistics.

Note: Logical and Relational models must be prepared and submitted separately.

LLO # 04: Design, implement, and query NoSQL databases to manage and retrieve non-relational data efficien

Question # 02: **[Weightage: 10 | Marks: 20 | Time: 25 minutes]**

The Smart City River Pollution and Water Quality Control System begins by checking available databases and then switches to a dedicated working database for operations. A collection is created to store river monitoring data. The system records detail of a newly appointed environmental officer and then stores multiple water sample readings collected from different river zones. All stored data is displayed to verify successful insertion. The system retrieves records from the North River Zone and arranges them in descending order of pollution level to highlight critical areas. It also filters out records that do not belong to the South Zone for comparative analysis. After monitoring, a specific polluted sample is updated as treated and reassigned to the Central Treatment Unit. The system then rechecks the updated records to ensure accuracy. For reporting, only selected fields are displayed, and finally, an analytical summary is generated by grouping pollution levels and calculating average contamination for decision-making.

Collection Name: water_quality

Attributes:

- sample_id
- zone
- pollution_level
- contamination_type
- status
- officer_id
- officer_name
- designation

Note: Students must identify and write the appropriate MongoDB commands used in the scenario.

LLO # 03: Apply advanced SQL and PL/SQL features to manage, automate, and enhance database operations.

Question # 03: Use HR **[Weightage: 10 | Marks: 20 | Time: 25 minutes]**

1. Using a **subquery**, identify employees whose salary is higher than the maximum salary of employees in other departments except their own. Also display their job title and department location by joining relevant HR tables for full context.
2. Create a structured output that maps employees with their job roles and department details, ensuring only those employees are shown whose job assignment is active and department is located in an approved region using join operations.
3. Using a **correlated subquery**, retrieve employees whose salary is greater than at least one other employee in the same department but less than the highest salary in that department, ensuring relative salary positioning within each department.
4. Perform a department-wise analysis by selecting employees whose salary lies within a defined range (above minimum but below maximum salary of their department), while also displaying department name and region using join operations.
5. The organization needs to control employee transfers between departments. Whenever an employee is moved, the system must ensure that the new department exists in the HR system and is active. If valid, it must record employee ID, old department, new department, and transfer date into a history table. It must also prevent transfers to inactive departments or invalid department codes.

...# 03: Apply advanced SQL and PL/SQL features to manage, automate, and enhance database ope...

...estion # 04: **[Weightage: 20 | Marks: 40 | Time: 40 minu...**

A: The system manages patient records and treatment activities in a hospital using only two tables:

- **Patients (patient_id, name, status, bill_amount)**
- **Treatment_Log (log_id, patient_id, activity, log_date)**

1. A new patient is being admitted into the hospital and their treatment entry must also be recorded in the log system. The system must ensure that both operations are completed together; if any issue occurs, no record should remain stored.
2. The hospital plans to change the status of all patients marked as "under observation" to "critical review". If the number of affected patients exceeds a safe operational limit, all changes must be undone.
3. Before adding a new patient record, the system must verify that no existing patient has the same name. If a duplicate is found, the entire process must be cancelled.
4. The hospital performs multiple actions together: a new patient is admitted, an existing patient's billing amount is updated, and a discharged patient record is removed. These actions must either all succeed or all fail together.
5. During a batch update of patient treatment status, the system defines a checkpoint after completing the first phase of updates. If an error occurs in the next phase, only the changes after that checkpoint should be reversed while keeping earlier valid changes intact.

Part B:

1. Write a stored procedure that calculates total and average salary for each department for the current year. The procedure should display department ID, total salary, and average salary.
2. Write a PL/SQL block that checks all employees and identifies those whose hire_date falls on a weekend (Saturday/Sunday). Insert these records into a separate table named weekend_hires_log.
3. Create an AFTER UPDATE trigger on the salary column of employees table. The trigger should store employee ID, old salary, new salary, and update date into a salary_audit table whenever salary is changed.
4. Write a stored procedure that identifies duplicate employees based on same first_name and last_name and removes duplicate records while keeping only one valid entry.
5. Write a PL/SQL block that loops through employees and assigns bonuses based on job_id:
   - Managers → 20% bonus
   - IT staff → 15% bonus
   - Others → 10% bonus

Must Return Your Paper after Completion!

Best of luck!